Nth Fibonacci

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The problem

- F(1)=0, F(2)=1.
- Iterative with two variables.

Intuition

- Each number only depends on the previous two, so we don't need recursion or an array.

Approach

- 1. Handle n of one and two
- 2. Keep the last two values
- 3. Slide them forward n minus two times
- 4. Return the latest

Java solution

```java
public class NthFibonacci {
    public static int getNthFib(int n) {
        int a = 0, b = 1;
        if (n == 1) return 0;
        for (int i = 3; i <= n; i++) { int c = a + b; a = b; b = c; }
        return b;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Each number only depends on the previous two, so we don't need recursion or an array.
- Complexity: O(n)